Problem Clarification

Friday, 27 October 2023 9:58 AM

$$\underline{F} = \iint_{@surface} (\hat{n} \cdot \underline{\underline{\Pi}}) \, dA$$

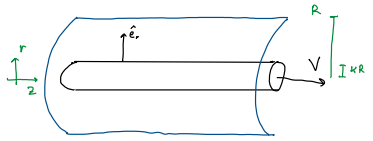
- $\hat{n}$: normal to surface
- $\underline{F}$: Force on a surface of area given by dA
- $\underline{\underline{\Pi}}$: Total stress tensor contains information about the stress at a point in a fluid

[Force on a surface (by the fluid flow) is positive]

∴ 1) Bounds on integral must define the area of the surface (not the fluid!)

2) The integral is evaluated @ the surface coordinates (i.e. the jacobian will be at specific coordinate)

eg:



+ Force on wire:

$$\underline{F} = \iint_{cs} (\hat{n} \cdot \underline{\underline{\Pi}}) \, dS$$

surface is the entire wire / small cylinder

where:

$$0 < \theta < 2\pi$$
$$0 < z < L$$

$$\therefore \underline{F} = \int_0^L \int_0^{2\pi} \left[\hat{e}_r \cdot \begin{pmatrix} -p & 0 & \mu \frac{\partial v_z}{\partial r} \\ 0 & -p & 0 \\ \mu \frac{\partial v_z}{\partial r} & 0 & -p \end{pmatrix} r \right]_{@r=KR} d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} \begin{pmatrix} -p \\ 0 \\ \mu \frac{\partial v_z}{\partial r} \end{pmatrix}_{r\theta z} \Bigg|_{@r=KR} KR \, d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} \left(-p \, \hat{e}_r + \mu \frac{\partial v_z}{\partial r} \hat{e}_z \right) \Big|_{@r=KR} KR \, d\theta \, dz$$

$$= \int_0^L \int_0^{2\pi} (-p \, \hat{e}_r) \Big|_{@r=KR} KR \, d\theta \, dz + \int_0^L \int_0^{2\pi} \mu \frac{\partial v_z}{\partial r} \Big|_{@r=KR} \hat{e}_z \, KR \, d\theta \, dz$$

$$v_z = V \left[\frac{\ln(r/R)}{\ln(K)} \right] \rightarrow \text{solved from previous part}$$

$$\frac{\partial v_z}{\partial r} = \frac{V}{\ln K} \left(\frac{R}{r} \right) \cdot \frac{1}{R} = \frac{V}{\ln K} \frac{1}{r}$$

↳ @ $r = KR$ → $\frac{\partial v_z}{\partial r} = \frac{V}{\ln K} \left(\frac{1}{KR} \right) \neq f(r, \theta, z)$

$$\therefore (KRL) \int_0^{2\pi} (-p \, \hat{e}_r) \Big|_{@r=KR} d\theta + L \int_0^{2\pi} \left(\frac{\mu V}{\ln K} \cdot \frac{1}{KR} \cdot KR \right) \hat{e}_z \, d\theta$$

$\hat{e}_r$ → unit vector but has some θ dependence

↳ express in θ units : $\hat{e}_r = \cos\theta \, \hat{e}_x + \sin\theta \, \hat{e}_y$

$$= (KRL) \int_0^{2\pi} -p_{KR} \cdot \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}_{xyz} d\theta + \frac{\mu V L}{\ln K} (2\pi) \, \hat{e}_z$$

$\int_0^{2\pi} \cos\theta \, d\theta = \int_0^{2\pi} \sin\theta = 0$; $\hat{e}_z$ independent of θ direction

Centre term goes to zero!)

$$\underline{F} = \frac{\mu V L (2\pi)}{\ln K} \hat{e}_z = \begin{pmatrix} 0 \\ 0 \\ \end{pmatrix}$$

centre term goes to zero :)

$$\underline{F} = \frac{\mu V L (2\pi)}{\ln k} \hat{e}_z = \begin{pmatrix} 0 \\ 0 \\ \frac{\mu V L (2\pi)}{\ln k} \end{pmatrix}_{r\theta z}$$

→ Always a vector!

→ physically: + longer the wire, longer the force required to yank it out
+ Faster the V, more force exerted on the wire surface

Makes sense!